

SAFETY DATA SHEET

1. Chemical and company identification

Name of chemical (Product name)	2-HEMA(2-Hydroxyethyl Methacrylate)
Supplier's company name, address and phone number	
Company name	MITSUBISHI GAS CHEMICAL COMPANY, INC.
Address	5-2, Marunouch 2-chome, Chiyoda-ku, Tokyo 100-8324, Japan
Department in Charge	Basic Chemicals Division I, Basic Chemicals Business Sector
Telephone Number	+81-3-3283-4788
Facsimile Number	+81-3-3214-0930
Email Address	msdsnc@mgc.co.jp
Emergency Telephone Number	+81-3-6890-8677(Verisk 3E)
Access Code	335392
Product code	1-34-0800-2

2. Hazards identification

GHS classification

Physical hazards	Flammable liquids	Not classified
Health hazards	Acute toxicity, oral	Not classified
	Acute toxicity, dermal	Not classified
	Skin corrosion/irritation	Not classified
	Serious eye damage/eye irritation	Category 2A
	Sensitization, skin	Category 1
	Germ cell mutagenicity	Not classified
Environmental hazards	Short-term (acute) hazardous to the aquatic environment	Not classified
	Long-term (chronic) hazardous to the aquatic environment	Not classified

GHS label elements

Pictograms



Signal words

Warning

Hazard statement

H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.

Precautionary statement

Prevention

P261	Avoid breathing mist/vapors.
P264	Wash thoroughly after handling.
P272	Contaminated work clothing should not be allowed out of the workplace.
P280	Wear eye protection/face protection.
P280	Wear protective gloves.

Response

P302 + P352	IF ON SKIN: Wash with plenty of water.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P333 + P313	If skin irritation or rash occurs: Get medical advice/attention.
P337 + P313	If eye irritation persists: Get medical advice/attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Storage

Store away from incompatible materials.

Disposal

P501	Dispose of contents/container in accordance with local/regional/national/international regulations.
Other hazards which do not result in classification	None known.
Supplemental information	None.
Main symptoms and emergency overview	
Main symptoms	Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. May cause an allergic skin reaction. Dermatitis. Rash.
Emergency overview	Causes serious eye irritation. May cause an allergic skin reaction.

3. Composition/information on ingredients

Substance or mixture	Substance			
Chemical name or generic name	CAS Number	Gazette notification		
		ENCS no.	ISHL no.	Concentration (%)
2-Hydroxyethyl Methacrylate	868-77-9	(2)-1044	(2)-1044	≥ 97

4. First aid measures

If inhaled	Move to fresh air. Call a physician if symptoms develop or persist.
If on skin	Remove contaminated clothing immediately and wash skin with soap and water. In case of eczema or other skin disorders: Seek medical attention and take along these instructions.
If in eyes	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
If swallowed	Rinse mouth. Get medical attention if symptoms occur.
Most important symptoms/effects, acute and delayed	Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. May cause an allergic skin reaction. Dermatitis. Rash.
Protection of first-aid responders	Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves. Wash contaminated clothing before reuse.
Notes to physician	Provide general supportive measures and treat symptomatically. Keep victim under observation. Symptoms may be delayed.

5. Fire-fighting measures

Extinguishing media	Water fog. Foam. Dry chemical powder. Carbon dioxide (CO ₂).
Extinguishing media to avoid	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards	During fire, gases hazardous to health may be formed.
Special fire fighting procedures	Move containers from fire area if you can do so without risk.
Protection of fire-fighters	Self-contained breathing apparatus and full protective clothing must be worn in case of fire.
General fire hazards	No unusual fire or explosion hazards noted.
Specific methods	Use standard firefighting procedures and consider the hazards of other involved materials.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Wear appropriate protective equipment and clothing during clean-up. Avoid breathing mist/vapors. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Environmental precautions	Avoid discharge into drains, water courses or onto the ground.
Methods and materials for containment and cleaning up	This product is miscible in water. Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Absorb in vermiculite, dry sand or earth and place into containers. Following product recovery, flush area with water. Small Spills: Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. Never return spills to original containers for re-use. For waste disposal, see section 13 of the SDS.

7. Handling and storage

Handling	
Technical measures (e.g. Local and general ventilation)	Provide adequate ventilation.

Safe handling advice	Avoid breathing mist/vapors. Avoid contact with eyes, skin, and clothing. Observe good industrial hygiene practices. Use personal protection recommended in Section 8 of the SDS.
Contact avoidance measures	For further information, please refer to section 10 of the SDS.
Hygiene measures	Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Contaminated work clothing should not be allowed out of the workplace.
Storage	
Safe storage conditions	Keep container tightly closed. Store away from incompatible materials (see Section 10 of the SDS).
Safe packaging materials	Store in original tightly closed container.

8. Exposure controls/personal protection

Control parameters	Follow standard monitoring procedures.
Occupational exposure limits	No exposure limits noted for ingredient(s).
Engineering measures	Good general ventilation should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station.
Personal protective equipment	
Respiratory protection	In case of insufficient ventilation, wear suitable respiratory equipment.
Hand protection	Wear appropriate chemical resistant gloves.
Eye protection	Wear safety glasses with side shields (or goggles). Face shield is recommended.
Skin and body protection	Wear appropriate chemical resistant clothing. Use of an impervious apron is recommended.

9. Physical and chemical properties

Physical state	Liquid.
Form	Liquid.
Color	Colorless.
Odor	Aromatic.
Melting point/freezing point	10.4 °F (-12 °C)
Boiling point, initial boiling point, and boiling range	401 °F (205 °C)
Combustibility	Not applicable.
Lower and upper explosion limit / flammability limit	
Flammability limit - lower (%)	None known
Flammability limit - upper (%)	None known
Explosive limit - lower (%)	None known
Explosive limit - upper (%)	None known
Flash point	224.60 °F (107.00 °C)
Auto-ignition temperature	None known
Decomposition temperature	None known
pH	None known
Kinematic viscosity	Not available.
Solubility(ies)	
Solubility (water)	Soluble in water.
Partition coefficient (n-octanol/water) (log value)	log Pow = 0.42 (25°C)
Vapor pressure	None known
Density and/or relative density	
Density	Not available.
Relative density	Not available.
Vapor density	None known
Particle characteristics	Not available.

Other information

Explosive properties	Not explosive.
Oxidizing properties	Not oxidizing.
Specific gravity	1.072 - 1.076 (20° C)

10. Stability and reactivity

Reactivity	The product is stable and non-reactive under normal conditions of use, storage and transport.
Chemical stability	Material is stable under normal conditions.
Possibility of hazardous reactions	Polymerizes in sunlight, heat, catalysts, etc.
Conditions to avoid	Direct sunlight, high temperature, contamination with impurities, etc.
Incompatible materials	None known.
Hazardous decomposition products	None known.

11. Toxicological information

Acute toxicity	Not available.
Skin corrosion/irritation	Based on available data, the classification criteria are not met.
Serious eye damage/eye irritation	Causes serious eye irritation.
Respiratory or skin sensitization	
Respiratory sensitization	Due to partial or complete lack of data the classification is not possible.
Skin sensitization	May cause an allergic skin reaction.
Germ cell mutagenicity	Based on available data, the classification criteria are not met.
Carcinogenicity	Due to partial or complete lack of data the classification is not possible.
Reproductive toxicity	Due to partial or complete lack of data the classification is not possible.
Specific target organ toxicity - single exposure	Due to partial or complete lack of data the classification is not possible.
Specific target organ toxicity - repeated exposure	Due to partial or complete lack of data the classification is not possible.
Aspiration hazard	Due to partial or complete lack of data the classification is not possible.

12. Ecological information

Ecotoxicological data

Components	Species	Test Results
2-Hydroxyethyl Methacrylate (CAS 868-77-9)		
Aquatic		
Acute		
Fish	LC50	Fathead minnow (Pimephales promelas) 213 - 242 mg/l, 96 hours
Ecotoxicity		The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.
Persistence and degradability		No data is available on the degradability of any ingredients in the mixture.
Bioaccumulation		
Bioaccumulative potential		
Octanol/water partition coefficient log Kow		
2-Hydroxyethyl Methacrylate	0.47	
Mobility in soil		Not available.
Hazardous to the ozone layer		No data available.
Other hazardous effects		No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

13. Disposal considerations

Residual waste	Dispose of in accordance with local regulations. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe manner.
Contaminated packaging	Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

Local disposal regulations

Contract with a disposal operator licensed by the Law on Disposal and Cleaning. Dispose of contents/container in accordance with local/regional/national/international regulations. When your own wastewater treatment plant is not available, collect entire waste and then charge to a licensed industrial waste management professional with manifests for industrial waste.

14. Transport information**IATA**

Not regulated as dangerous goods.

IMDG

Not regulated as dangerous goods.

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code Not established.

National regulations Follow regulation in section 15 for domestic transportation.

15. Regulatory information**Industrial Safety and Health Act****Notifiable substances**

Not regulated.

Labeling substances

Not regulated.

Poisonous and Deleterious Substances Control Act**Specified poisonous substances**

Not regulated.

Poisonous substances

Not regulated.

Deleterious substances

Not regulated.

Act on the Regulation of Manufacture and Evaluation of Chemical Substances**Class I specified chemical substances**

Not regulated.

Class II specified chemical substances

Not regulated.

Monitoring chemical substances

Not regulated.

Priority Assessment Chemical Substances (PACs)

Not regulated.

Reporting Exempted Substances

Not regulated.

Law concerning Pollutant Release and Transfer Register**Specified class 1 substances (substance name, ordinance number and content)**

Not regulated.

Class 1 substances (substance name, ordinance number and content)

Not regulated.

Class 2 substances (substance name, ordinance number and content)

Not regulated.

Fire Service Act

Class 4 Group 3 oils (Water soluble) Hazard rank III (Storage limit: 4000 l)

Ship Safety Law, Dangerous Goods Marine Transport and Storage Rule

Not regulated.

Air Law, Enforcement Rule

Not regulated.

Explosives Control Act

Not regulated.

16. Other information**Bibliography**

ACGIH Documentation of the Threshold Limit Values and Biological Exposure Indices
Japan Chemical Industry Association (JCIA) GHS Guideline, June 2019
HSDB® - Hazardous Substances Data Bank
IARC Monographs. Overall Evaluation of Carcinogenicity
Japan Society for Occupational Health, Recommendation of Occupational Exposure Limits
JIS Z 7252:2019 Classification of chemicals based on "Globally Harmonized System of Classification and Labelling of Chemicals (GHS)"
JIS Z 7253:2019 Hazard communication of chemicals based on GHS - Labelling and Safety Data Sheet (SDS)
National Toxicology Program (NTP) Report on Carcinogens

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